

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AAE5206
Subject Title	Artificial Intelligence in Aerospace Engineering
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	The objectives of this subject are to: 1. introduce the principles, concepts and models of most popular deep learning algorithms used in aerospace research and industry. 2. provide both theoretical and practical understanding of machine learning models such as deep neural networks. 3. develop proficiency in designing, training, and optimizing machine learning models using Python and PyTorch.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. understand the fundamental principles and algorithms of artificial intelligence models. b. develop insights into the effectiveness and applicability of artificial intelligence techniques in the aerospace industry. c. gain knowledge of applying artificial intelligence techniques to various cutting-edge aerospace applications. d. design innovative artificial intelligence solutions to address complex challenges in aerospace engineering, considering various requirements and constraints. e. participate effectively in teamwork, presentation, and technical writing, enhancing collaboration and communication skills.
Subject Synopsis/ Indicative Syllabus	Foundations of Mathematics and Programming for Aerospace AI: Introduction to the mathematical and coding essentials for aerospace AI. Topics include linear algebra, probability theory, optimization methods, hands-on Python and PyTorch exercises. Emphasis on translating theoretical concepts into functional models, fostering confidence in implementing algorithms for aerospace challenges. Classical Supervised Algorithms for Aerospace Data Analysis: Exploration of classical supervised learning techniques for aerospace data analysis. Includes concept learning, regression and classification algorithms (SVM, k-NN, decision trees, and naïve Bayes), with hands-on implementation. Developing proficiency in algorithm selection and performance analysis for aerospace data interpretation.

	Deep Spatial Learning in Aerospace Applications: Introduces deep learning architectures for extracting and interpreting spatial information in aerospace contexts. Covers multilayer perceptrons for feature embedding, convolutional neural networks for satellite imagery and grid-based analysis, and graph neural networks for unstructured data processing. Emphasis on leveraging spatial correlations to enhance pattern recognition and mapping fidelity. Temporal Sequence Modeling for Aerospace Systems: Exploration of advanced neural techniques for capturing temporal relationships in aerospace sensor data. Topics include recurrent neural networks, LSTM and GRU cells, and attention-based models, applied to control-signal forecasting and anomaly detection. Discussion on modeling time-dependent patterns to support accurate trajectory prediction, real-time monitoring, and adaptive control strategies. Data Analytics and Machine Learning Applications in Aerospace: Application of data analytics and machine learning in aerospace. Covers pattern discovery, topic modeling, genomics, and prediction in aerospace contexts. Discussion on scalability, interpretability, and legal/social/ethical considerations in aerospace data analytics.																																																						
Teaching/Learning Methodology	The course facilitates learning through a combination of lectures, tutorials, lab sessions, and a group project, and does not include a reading week. Lectures provide comprehensive coverage of the course concepts, supplemented by examples and interactive question & answer sessions for clarity. Tutorials and lab sessions reinforce theoretical knowledge with practical exercises, emphasizing hands-on experience with machine learning tools and techniques. The group project engages students in collaborative work to apply theoretical concepts to real-world scenarios, enhancing their analytical and problem-solving skills.																																																						
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="5">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>1. Assignments</td><td>30%</td><td>√</td><td>√</td><td>√</td><td>√</td><td></td></tr><tr><td>2. Quizzes</td><td>30%</td><td>√</td><td>√</td><td>√</td><td>√</td><td></td></tr><tr><td>3. Application Development</td><td>20%</td><td>√</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>4. Project Report</td><td>10%</td><td>√</td><td>√</td><td>√</td><td></td><td>√</td></tr><tr><td>5. Project Presentation</td><td>10%</td><td>√</td><td>√</td><td>√</td><td></td><td>√</td></tr><tr><td>Total</td><td>100 %</td><td colspan="5"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					a	b	c	d	e	1. Assignments	30%	√	√	√	√		2. Quizzes	30%	√	√	√	√		3. Application Development	20%	√	√	√	√	√	4. Project Report	10%	√	√	√		√	5. Project Presentation	10%	√	√	√		√	Total	100 %					
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	Continuous assessments in this course consist of assignments, quizzes, application development, project reports, and project presentations, each designed to facilitate students in achieving the intended learning outcomes. Assignments are crafted to encourage students to acquire a good understanding of machine learning models and algorithms, providing opportunities for practice to enrich their hands-on experience with various software tools and ensuring they develop insights into the effectiveness of machine learning techniques. Quizzes are utilized to ensure that students grasp the essential concepts and can apply machine learning techniques to various cutting-edge applications, helping to assess their theoretical knowledge. Application Development is aimed at enhancing students' abilities to use different knowledge, principles, techniques, and tools to solve real-world problems in aerospace engineering, assessing their capacity to design machine learning solutions considering various requirements and issues. The Project Report allows students to demonstrate their understanding and application of machine learning techniques comprehensively, assessing their ability to analyze data, present findings, and critically evaluate their work. The Project Presentation assesses students' ability to communicate their work effectively, showcasing their teamwork, presentation, and technical writing skills, and provides an opportunity for them to demonstrate their analytical and problem-solving skills through system implementation and interpersonal communication. The continuous assessment evaluates students' overall understanding and usage of machine learning and data analytics techniques, ensuring they have a solid foundation in the subject matter.	
Student Study Effort Expected	Class contact:	
	▪ Lecture/Tests	26 Hrs.
	▪ Laboratory/Tutorial	13 Hrs.
	Other student study effort:	
	▪ Assignments, Projects, Quizzes	36 Hrs.
	▪ Literature Review, Self-study & Preparation	36 Hrs.
	Total student study effort	111 Hrs.
Reading List and References	1. Zhang A, Lipton ZC, Li M, Smola AJ. Dive into deep learning. Cambridge University Press, 2023. 2. Ian Goodfellow, Deep Learning, MIT, 2016. 3. Jared Dean, Big Data, Data Mining, and Machine Learning: Value Creation for Business Leaders and Practitioners. Wiley, 2014. 4. EMC Education Services (Editor), Data Science and Big Data Analytics: Discovering, Analyzing, Visualizing and Presenting Data, Wiley, 2015. 5. Christopher Bishop, Pattern Recognition and Machine Learning, Springer, 2006. 6. Deisenroth MP, Faisal AA, Ong CS. Mathematics for machine learning. Cambridge University Press; 2020 Apr 23. 7. Generative Artificial Intelligence Projects 2024 updated:	

	https://github.com/sindresorhus/awesome-chatgpt
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May 2024